

Credit Default Prediction

Objective

The goal of this project was to predict whether a credit card customer would default on their payment in the following month using historical financial and repayment information. Since missed repayments directly impact lending institutions through financial loss and increased risk exposure, the objective was to identify meaningful borrower patterns and build a model capable of supporting better credit risk assessment.

The project focused not only on predictive performance, but also on data quality, interpretability, and business relevance. Particular emphasis was placed on exploratory data analysis (EDA) and feature engineering, as repayment behaviour is often more informative than demographic information alone.

Approach

The project began with data quality assessment and preprocessing. The dataset was inspected for missing values, unusual categories, and potential inconsistencies. Special handling was applied to edge cases specified in the assignment, including:

- Merging EDUCATION values 0, 5, and 6 into an "Other" category
- Merging MARRIAGE = 0 into "Other"
- Explicitly checking negative BILL_AMT values, which likely represent overpayments or billing adjustments

Exploratory Data Analysis (EDA) was treated as a key part of the workflow. Distribution plots, repayment delay patterns, and default-rate comparisons across demographic groups were used to better understand borrower behaviour. Correlation analysis was also conducted to identify variables most associated with default.

Feature engineering focused on borrower financial behaviour rather than only static attributes. Three engineered features were created:

- **AVG_UTIL_RATE**: average credit utilization across six months
- **AVG_PAY_RATIO**: repayment consistency relative to billed amount
- **TOTAL_DELAY_MONTHS**: count of delayed repayment months

These features were designed to better capture repayment discipline and financial stress signals.

Model Choices and Tradeoffs

A **Logistic Regression** model was first implemented as a baseline because of its interpretability and simplicity. Since the dataset was moderately imbalanced (approximately 22% defaults), `class_weight='balanced'` was used to reduce bias toward the majority class.

To improve predictive performance, **XGBoost** was selected as the final tree-based model. Credit default behaviour is influenced by nonlinear repayment patterns and interactions between financial variables, making XGBoost more suitable than linear methods.

XGBoost outperformed Logistic Regression and achieved an **AUC-ROC score of approximately 0.75**, indicating reasonable predictive ability in distinguishing between default and non-default borrowers. Stratified 5-fold cross-validation was used to evaluate model consistency.

One tradeoff made was prioritizing model robustness and clean preprocessing over extensive hyperparameter tuning due to time constraints.

What I Would Improve Given More Time

Given additional time, several improvements could strengthen the project further.

First, I would implement **SHAP explainability** to better understand feature-level prediction behavior and improve model transparency for lending stakeholders.

Second, I would perform a **fairness analysis** to evaluate whether prediction performance differs across demographic groups such as education or gender.

Third, additional model tuning and experimentation (e.g., hyperparameter optimization for XGBoost) could potentially improve predictive performance beyond the current AUC score.

Finally, I would deploy the model through a lightweight **Streamlit interface**, allowing users to input borrower information and receive real-time default probability predictions.

Overall, the project reinforced an important insight:

Repayment behaviour appears to be a significantly stronger predictor of default risk than demographic characteristics alone.

This suggests that lending institutions may benefit from prioritizing behavioural financial signals when assessing borrower risk.